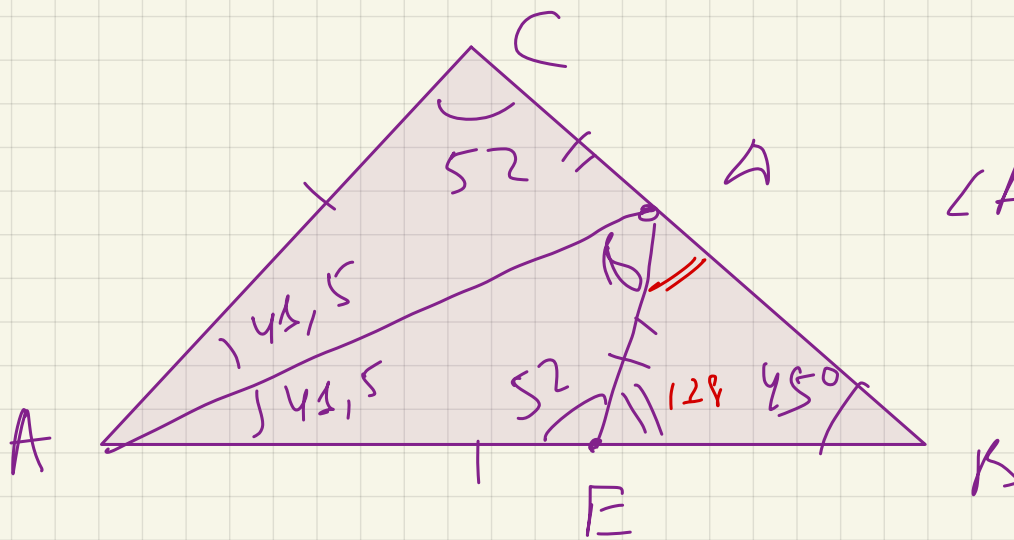






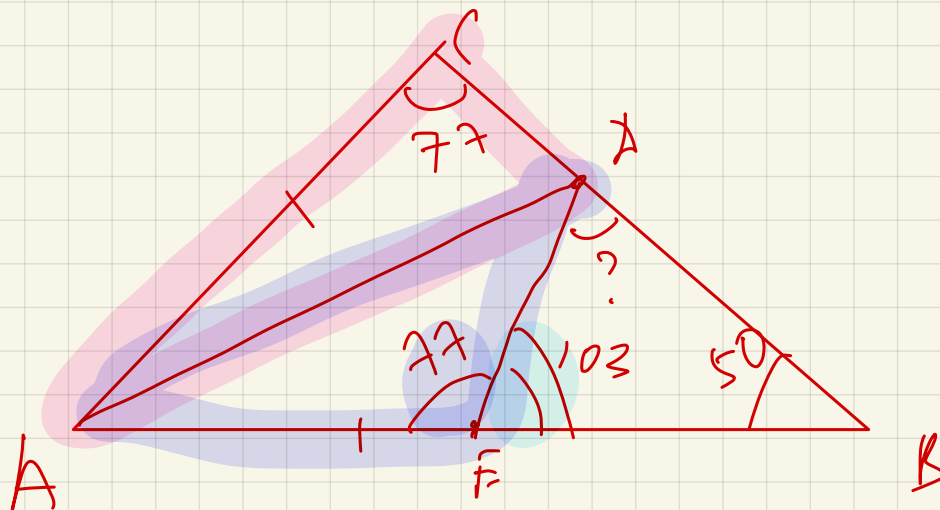
$$\angle BDE = ?$$

$$\angle A = 180 - 45 - 52$$

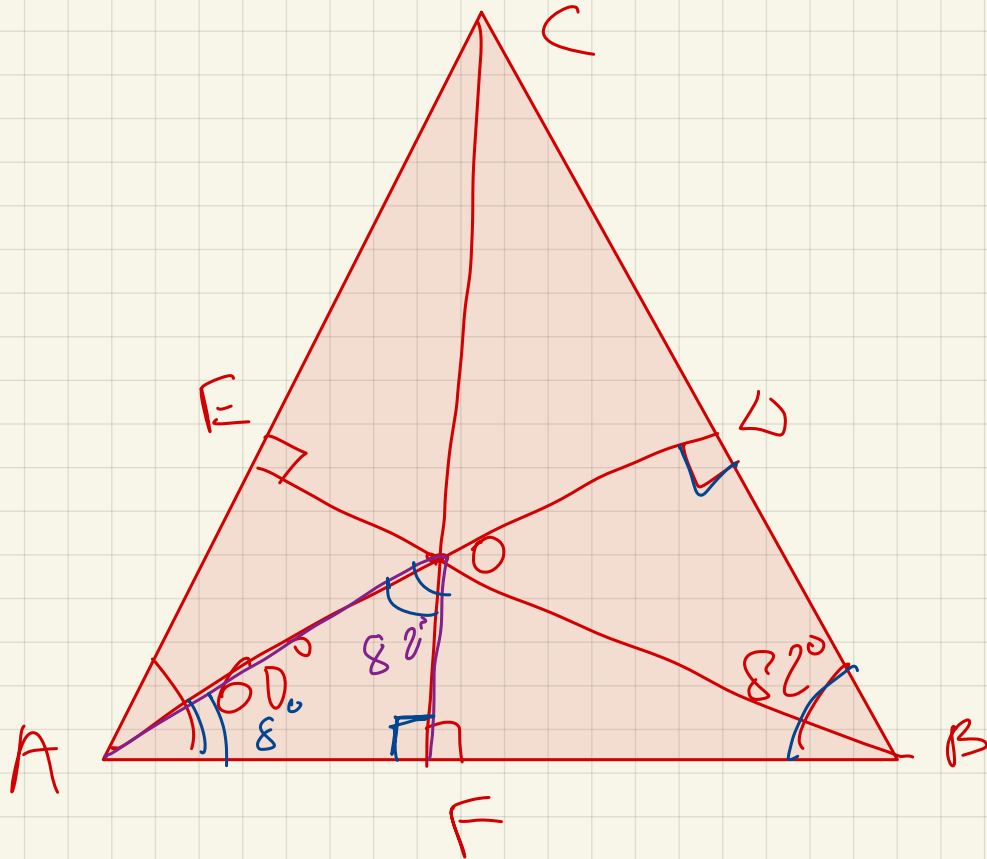
$$\angle AED = 180 - 52 = 128$$

$$\angle BDE = 180 - 128 - 45 = 7^\circ$$

В треугольнике ABC угол B равен 50° , угол C равен 77° , AD — биссектриса, E — такая точка на AB , что $AE = AC$. Найдите угол BDE . Ответ дайте в градусах.



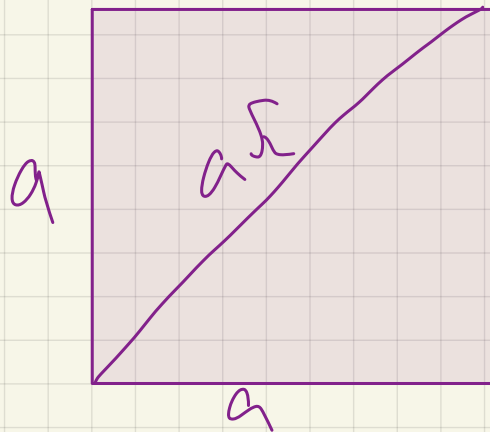
$$\angle AED = 103$$



$\angle AOF$?

$$\angle DAB = 180 - 90 - 82 = 8^\circ$$

Квадрат:

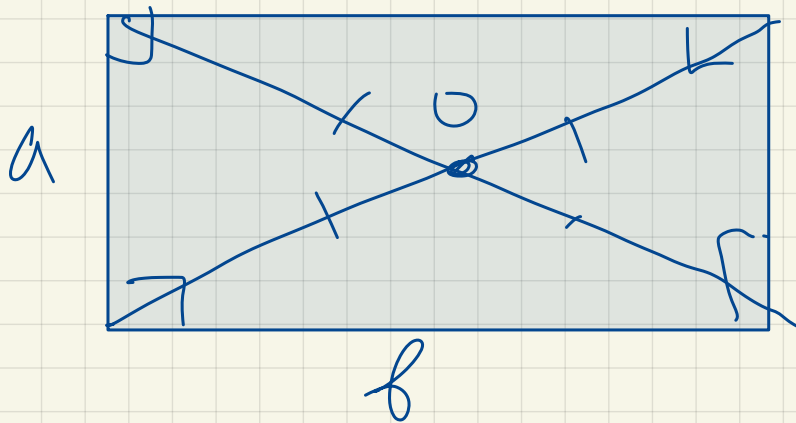


$$S = a^2$$

$$P = 4a$$

$$d = a\sqrt{2}$$

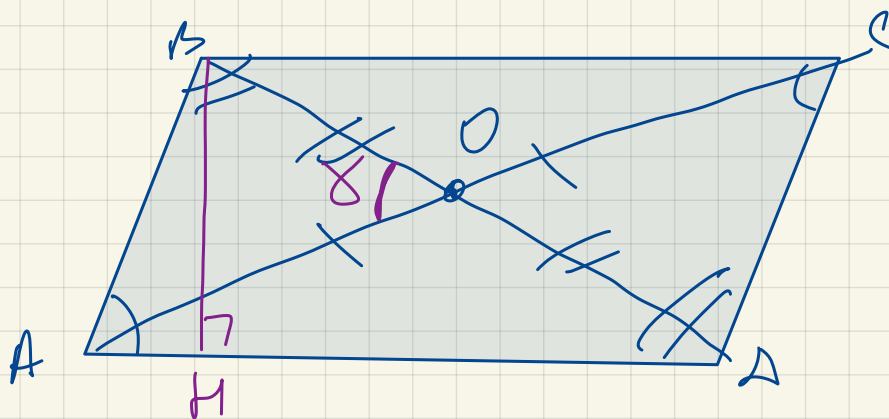
$\Gamma_{\text{параллелограмма}}:$



$$S = a \cdot b$$

$$P = 2a + 2b$$

$$d = \sqrt{a^2 + b^2}$$



$$1) \angle A = \angle C \quad \angle B = \angle D$$

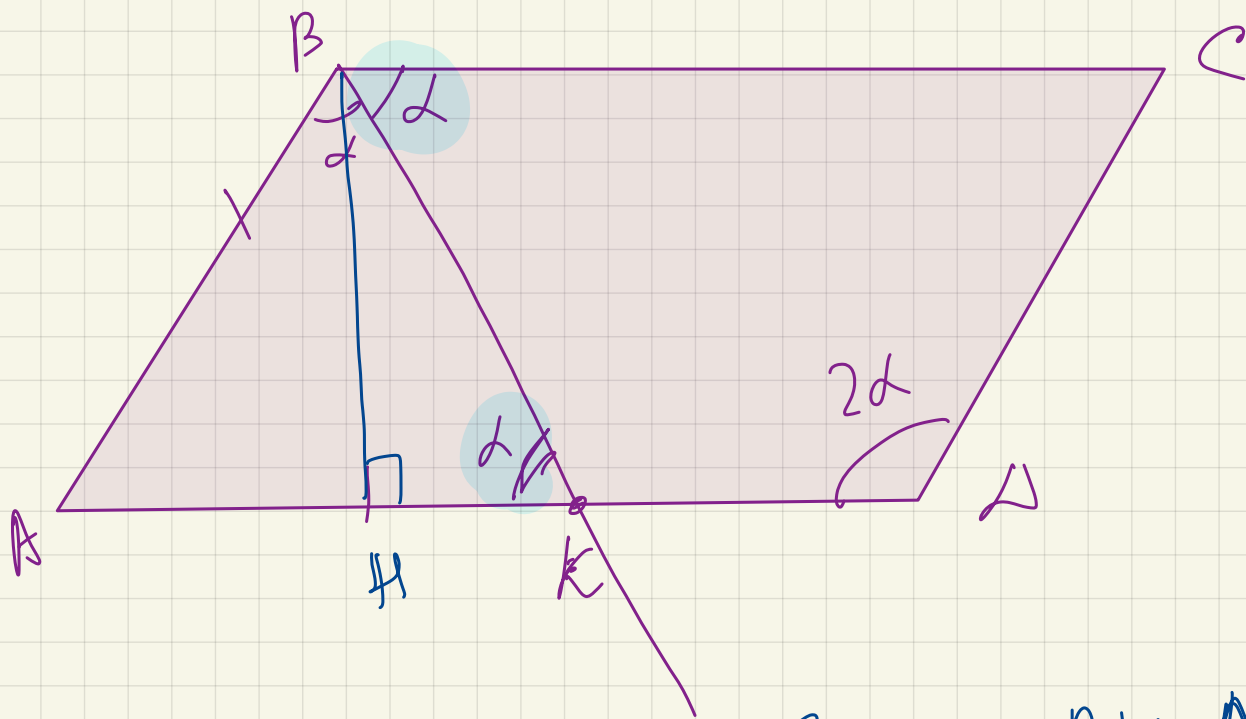
$$2) \angle A + \angle B = 180^\circ$$

3) O - т. пересечения диагоналей

$$S = BH \cdot AD$$

$$S = AB \cdot AD \cdot \sin A$$

$$S = \frac{d_1 \cdot d_2}{2} \cdot \sin \alpha$$



$$\frac{S_{ABK}}{S_{ABCD}} \quad ? \quad S_{ABK} = \frac{BH \cdot AK}{2}$$

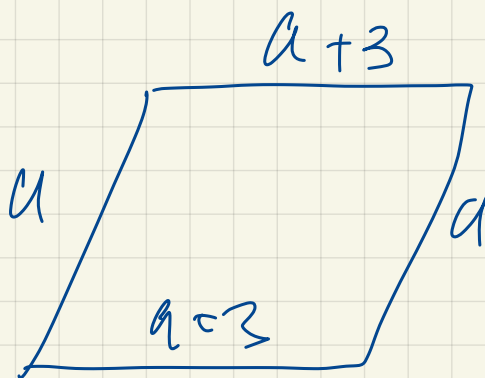
$$S_{ABCD} = BH \cdot AD$$

$$\frac{S_{ABK}}{S_{ABCD}} = \frac{\frac{BH \cdot AK}{2}}{BH \cdot AD} = \left(\frac{AK}{2AD} \right)$$

①

$$P = 46$$

a ?

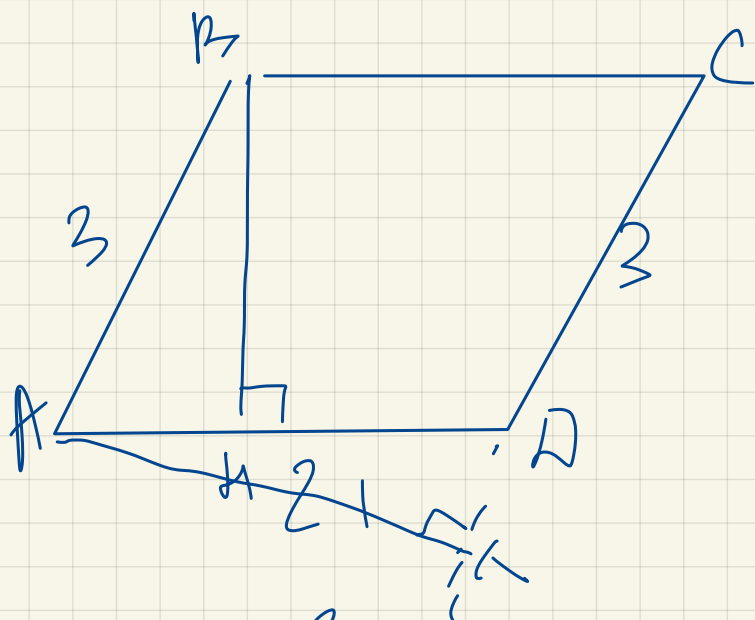


$$P = a + a + 3 + a + a + 3 = 4a + 6$$

$$4a + 6 = 46 \Rightarrow$$

$$4a = 40$$

$$a = 10$$



$$\sin A = \frac{6}{7}$$

$$\frac{BH}{AB} = \sin A = \frac{6}{7}$$

$$BH = \frac{18}{7}$$

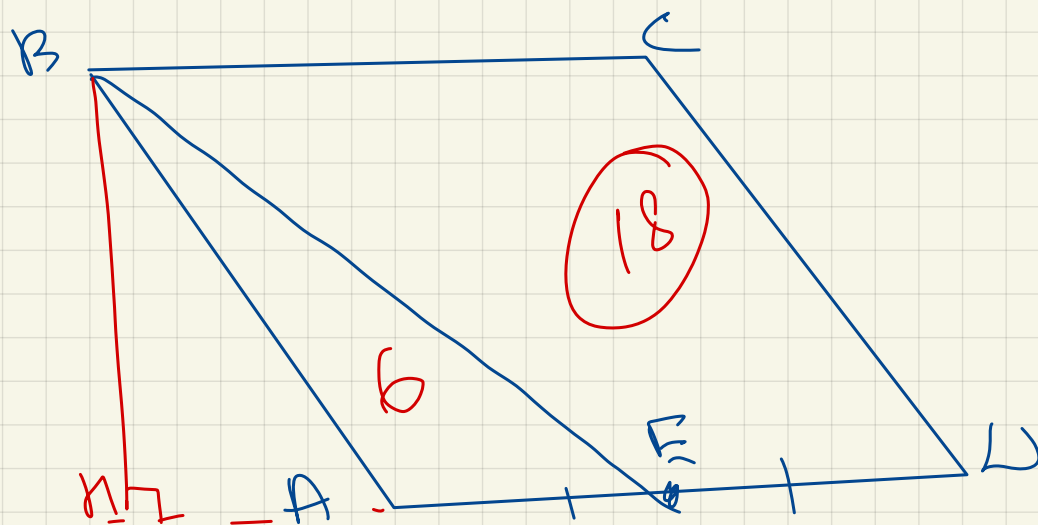
$$S = 8 \cdot 21 \cdot \frac{6}{7} = 54$$

$$AH \cdot CA = S$$

$$AH = \frac{54}{3} = 18$$

$$R.M. \cdot AH = 54$$

$$BH = \frac{54}{21}$$



$$S_{BENC}?$$

$$S_{ABCD} = 24$$

$$S_{ABE} = \frac{BH \cdot AE}{2}$$

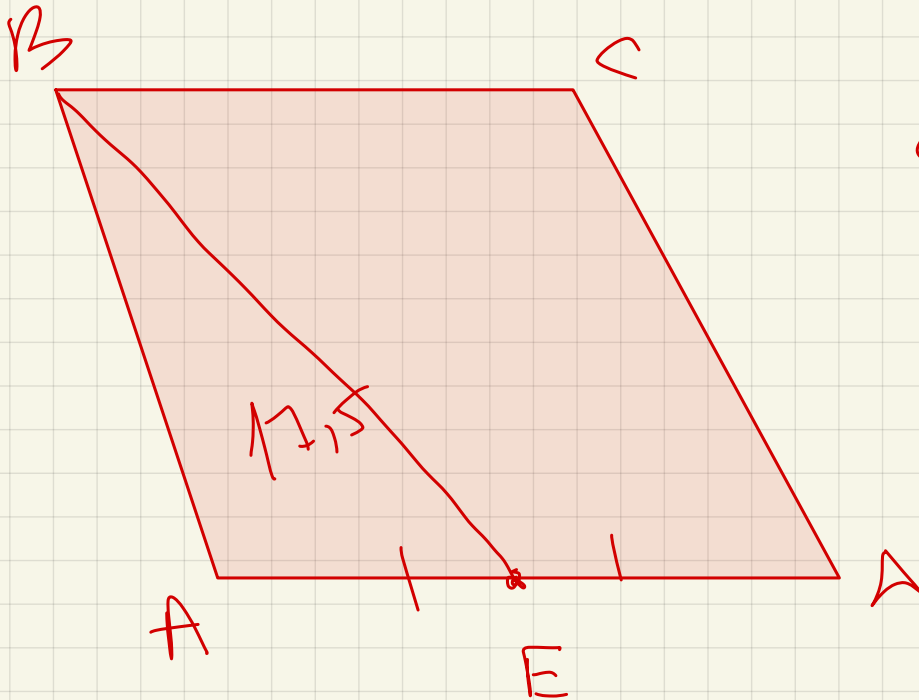
$$S_{ABCD} = BH \cdot AN$$

$$\frac{S_{ABE}}{S_{ABCD}} = \frac{\cancel{BH} \cdot AE}{2 \cancel{BH} AN}$$

$$= \frac{AE}{2AN} = \frac{AE}{2 \cdot 2AE}$$

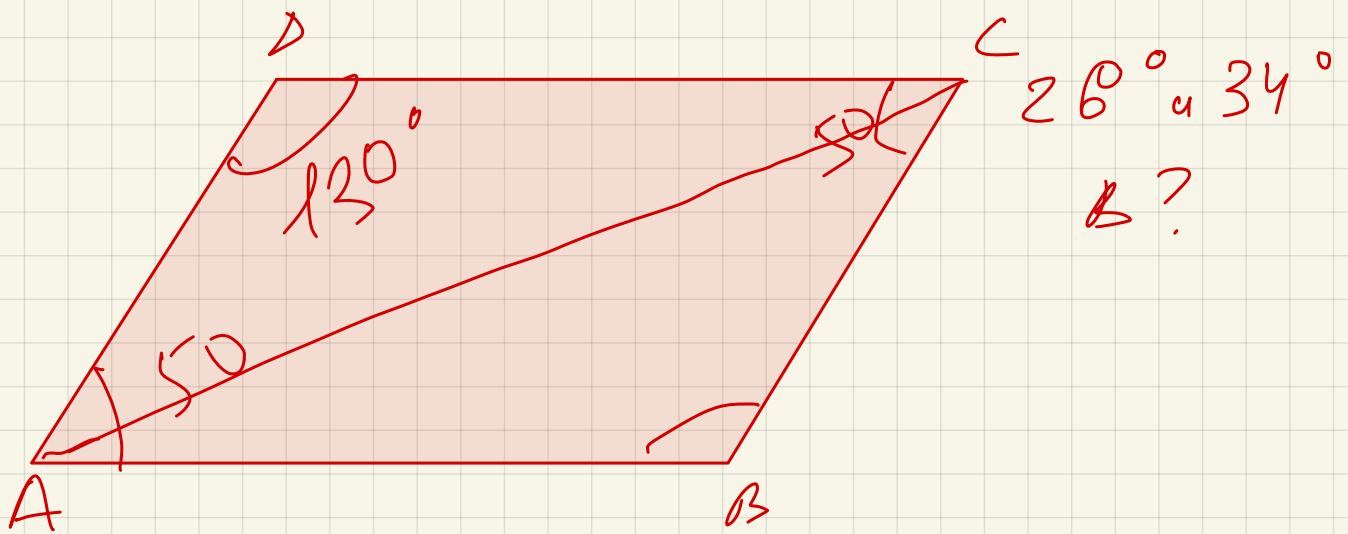
$$\frac{1}{4}$$

$$S_{ABE} = \frac{S_{ABCD}}{4} = 6$$

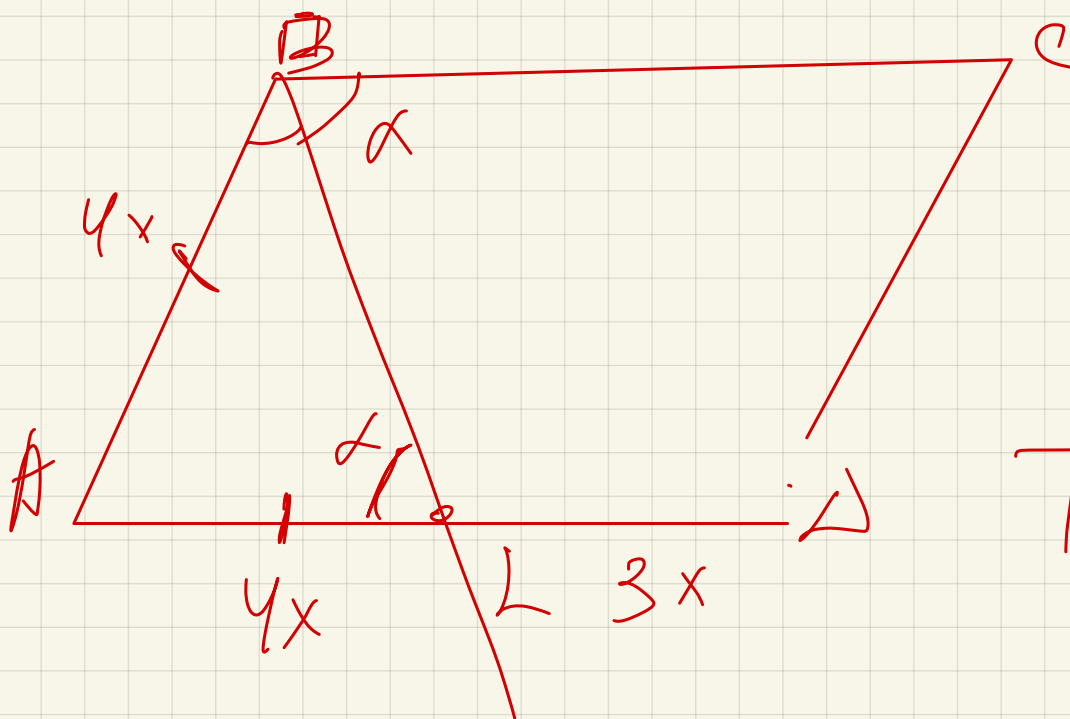


$$S_{ABCD} = 40$$

$$S_{BCDE}?$$



$$\sum_2 = 100^\circ \quad \angle ?$$



$$\frac{AL}{LD} = \frac{4x}{3x}$$

$$P = 88$$

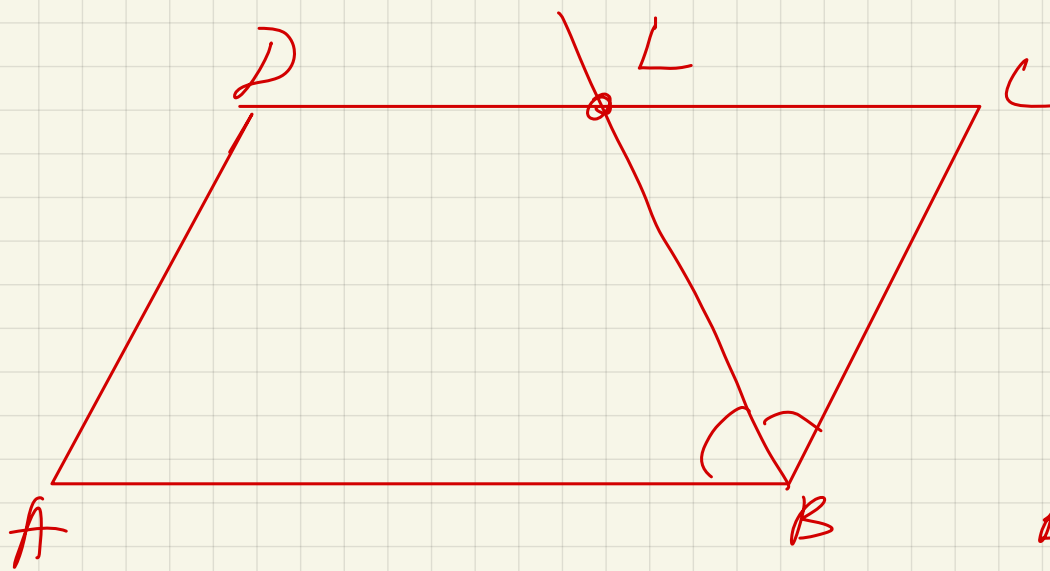
$$AD = ?$$

$$P = 4x + 7x + 4x + 7x = 22x$$

$$22x = 88$$

$$x = 4$$

$$AD = 28$$

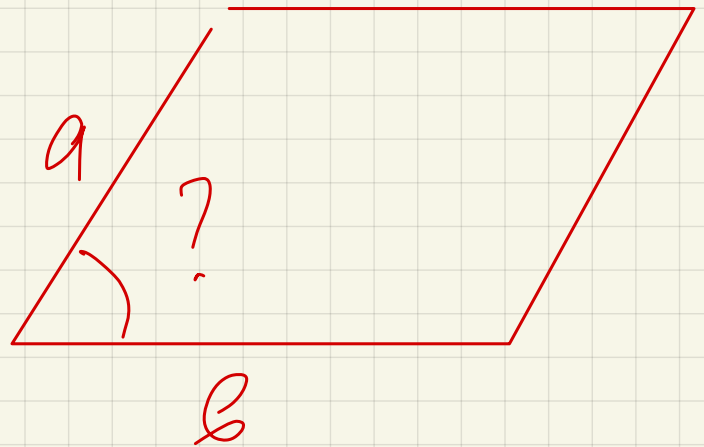
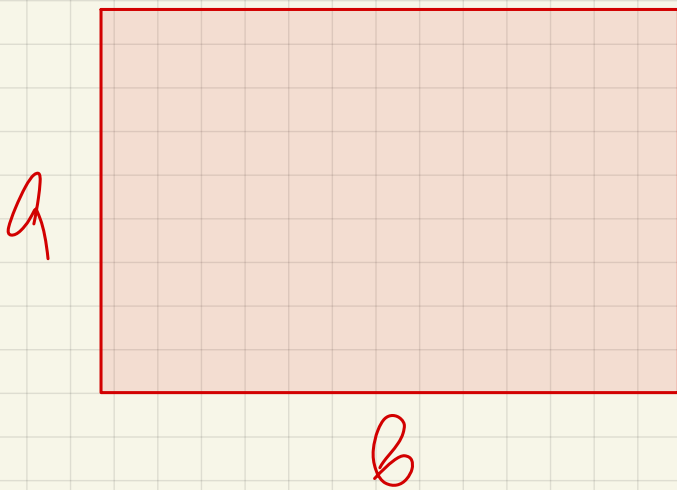


$$\frac{CL}{LD} = \frac{5}{8}$$

$$P = 108$$

BL - success

WC?



$$S_{\text{trap}} = \frac{S_{\text{trap}}}{2}$$

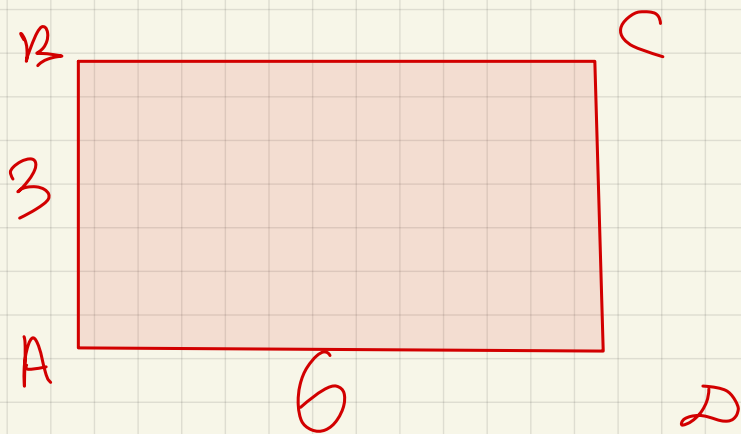
$$S_{\text{trap}} = ab$$

$$S_{\text{trap}} = a \cdot b \sin \alpha$$

$$ab \sin \alpha = \frac{ab}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$



P?

$$S = 18$$

$$\frac{AB}{AD} = \frac{1}{2}$$

$$AB = \frac{AD}{2}$$

$$AB \cdot AD = 18$$

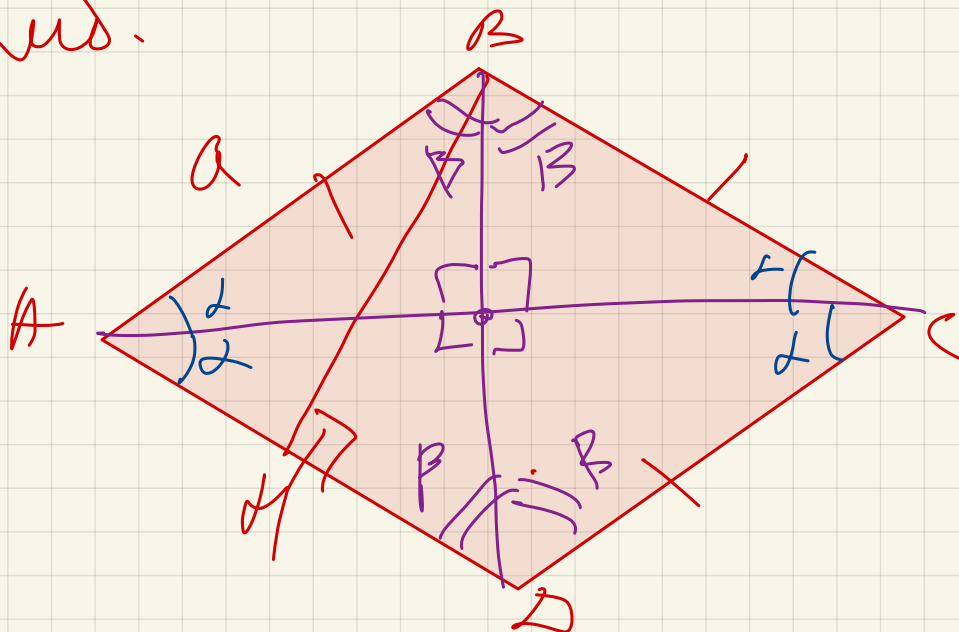
$$\frac{AD}{2} \cdot AD = 18$$

$$AD^2 = 36 \Rightarrow AD = 6$$

$$AB = \frac{6}{2} = 3$$

$$P = 18$$

Ровн.



$$\angle A = \angle C$$

$$\angle B = \angle D$$

|

$$S = a^2 \cdot \sin A$$

$$S = a \cdot BH$$

$$S = \frac{BH \cdot AC}{2}$$

$$d_1 \perp d_2$$

$$\sin 90^\circ = 1$$

$$\angle A + \angle B = 180^\circ$$